

Prayas 2.0 2027 (OG)

Physics

Motion in a Straight Line

DPP: 2

Total Duration - 25 Min

- Q1** A car travels half the distance with constant velocity of 40kmph and the remaining half with a constant velocity of 80kmph. The average velocity of the car is:
- (A) 32 km/hr
(B) 53.3 km/hr
(C) 43.2 km/hr
(D) 42 km/hr
- Q2** A scooter moves in a straight line. It covers one fourth distance with v_1 , two fourth with v_2 and remaining with speed v_3 . The average speed of scooter is
- (A) $\frac{4v_1v_2v_3}{v_1v_2+v_2v_3+2v_1v_3}$
(B) $\frac{v_1v_2v_3}{v_1+v_2+v_3}$
(C) $\frac{v_1v_2v_3}{v_1+2v_2+v_3}$
(D) $\frac{v_1+2v_2+v_3}{4}$
- Q3** A particle is moving along x -axis such that $x = 2 - 5t + 6t^2$. What is acceleration of the particle when its velocity is zero?
- (A) Zero
(B) 12 m/s^2
(C) -5 m/s^2
(D) $5/12 \text{ m/s}^2$
- Q4** A car is moving with a velocity of 20 m/s. The driver accelerated it for 10 seconds and reached a velocity of 40 m/s. What is the average acceleration?
- (A) 3 m/s^2
(B) 2 m/s^2
(C) 1 m/s^2
(D) zero
- Q5** A car is moving with a velocity of 30 m/s. The driver applied brake for 5 seconds to bring it down to zero. What is the average acceleration?
- (A) -5 m/s^2
(B) 6 m/s^2
(C) -6 m/s^2
(D) Zero
- Q6** An α -particle in a cyclotron changes its velocity from 30 km/s south to 40 km/s west in 10 second what is the magnitude of average acceleration during this time
- (A) 5 Km/s^2
(B) 7 Km/s^2
(C) 9 Km/s^2
(D) 11 Km/s^2
- Q7** A truck travels a distance A to B at a speed of 40 km/h and returns to A at a speed of 50 km/h, then what is the average velocity of the whole journey?
- (A) 34.5 km/h
(B) Zero
(C) 35 km/hr
(D) 40 km/hr
- Q8** A person travelling on a straight line moves with a uniform velocity v_1 for some distance and with uniform velocity v_2 for the next equal distance. The average velocity v is given by
- (A) $v = \frac{v_1+v_2}{2}$
(B) $v = \sqrt{v_1 v_2}$
(C) $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$
(D) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$



Q9 A car covers a distance of 2 km in 2.5 minute, if it covers half of the distance with speed 40 km/hr. the rest distance it will cover with speed

- (A) 56 km/hr
- (B) 60 km/hr
- (C) 50 km/hr
- (D) 48 km/hr

Q10 A bicyclist encounters a series of hills. Uphill speed is always v_1 and downhill speed is always v_2 . The total distance travelled is ℓ , with uphill and downhill portions of equal length. The cyclists average speed is

- (A) $\frac{v_1}{v_2}$
- (B) $\frac{v_2}{v_1}$
- (C) $\frac{v_1 v_2}{v_1 + v_2}$
- (D) $\frac{2v_1 v_2}{v_1 + v_2}$

Q11 A motor car covers $\frac{1}{3}$ rd part of total distance with $v_1 = 10$ km/hr, second $\frac{1}{3}$ rd part with $v_2 = 20$ km/hr and rest $\frac{1}{3}$ rd part with $v_3 = 60$ km/hr. What is the average speed of the car?

- (A) 18 km/hr
- (B) 45 km/hr
- (C) 6 km/hr
- (D) 22.5 km/hr

Q12 A particle moving in a straight line covers half the distance with speed of 12 m/s. The other half of the distance is covered in two equal time intervals with speed of 4.5 m/s and 7.5 m/s respectively. The average speed of the particle during this motion is:

- (A) 8.0 m/s
- (B) 12.0 m/s
- (C) 10.0 m/s
- (D) 9.8 m/s

Q13 A body covers one-third of the distance with a velocity v_1 , the second one-third of the distance with a velocity v_2 , and the last one-third of the

distance with a velocity v_3 . The average velocity is:

- (A) $\frac{v_1 + v_2 + v_3}{3}$
- (B) $\frac{3v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$
- (C) $\frac{v_1 v_2 + v_2 v_3 + v_3 v_1}{3}$
- (D) $\frac{v_1 v_2 v_3}{3}$

Q14 A car travels a distance d on a straight road in two hours and then returns to the starting point in next three hours. Its average speed is:

- (A) $\frac{d}{5}$
- (B) $\frac{2d}{5}$
- (C) $\frac{d}{2} + \frac{d}{3}$
- (D) none of these

Q15 A particle moves in the east direction with 15 m/sec for 2sec then northwards with 5 m/s for 8sec. Average speed of the particle is:

- (A) 1 m/s
- (B) 5 m/s
- (C) 7 m/s
- (D) 10 m/s



Answer Key

Q1 (B)
Q2 (A)
Q3 (B)
Q4 (B)
Q5 (C)
Q6 (A)
Q7 (B)
Q8 (C)

Q9 (B)
Q10 (D)
Q11 (A)
Q12 (A)
Q13 (B)
Q14 (B)
Q15 (C)



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